Week 1 - Probability

Handout: 17.02.2026 18:00
Due: 01.03.2026 15:00

Spring 2026

Stochastics and Machine Learning

</> The Birthday Paradox in Python

Birthday Paradox in Python

Code a Python simulation for the birthday problem, i.e. estimate the probability that at least nrepeats people among a group of npeople people share the same birthday, assuming a year has ndays days.

Your task

1. Analytical approach: Implement the function `analytical_prob(npeople: int, ndays: int)` that computes the analytical probability of finding a `nrepeats=2` match within npeople. Your computation should match the answer provided in the previous exercise. The function `math.perm(n, k)` from the math libary might be useful (see [math.perm (https://docs.python.org/3/library/math.html#math.perm)](https://docs.python.org/3/library/math.html#math.perm)).

2. Simulation: Implement the function `simulated_prob(npeople: int, nrepeats: int , ndays: int,  ntrials: int)` which estimates the birthday problem probability by simulating `ntrials` experiments. For each experiment, sample npeople birthdays uniformly at random (assuming each day is equally likely and that the year has ndays days). Check if at least `nrepeats` people share the same birthday. For this you can use the already implemented function `is_match`. Return the estimated probability by averaging the counts among number of experiments.

3. (*Only for nrepeats=2*) analyze how the probability of a match increases as a function of the the number of people, compare the plots using the analytical probability and the simulated one. Implement the function `graph_comparison` and do 2 plots: simulated_probability vs npeople and analytical_probability vs npeople for `npeople = 1 to 100`. Fix `ntrials = 1000` to get a good simulation, but you can also experiment with lower numbers.

Questions:

Analytical probability: What is the analytical probability for `npeople=23` and `nrepeats=2`?

Effect of ntrials: Try different values (e.g., 100, 1000, 10,000) for the number of simulations. How does increasing `ntrials` affect the estimated probability?

Finding the Critical npeople: Find the smallest number of people npeople for which the probability ≥ 0.97.

This task was inspired by the Introduction to Probability and Statistics Course S22 from MIT.